

A Human-Like Agent For Screen-Aware Task Automation



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Problem & Impact

- Manual UI testing is slow and error-prone
- Automation fails on legacy/black-box systems
- Delays, higher costs, and missed bugs

Proof of Concept

- Agent observes screen → plans → acts → evaluates
- Demonstrates AI + vision + automation in QA

Key Technologies

- OpenCV + Tesseract OCR
- LangChain + LLaMA via Ollama
- HID emulation

Who is Affected

- QA teams testing legacy, secure, or third-party systems
- Critical domains: banking, healthcare, enterprise IT

Risk Mitigation

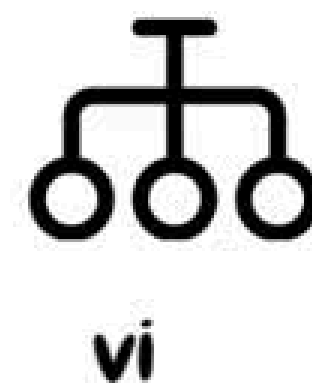
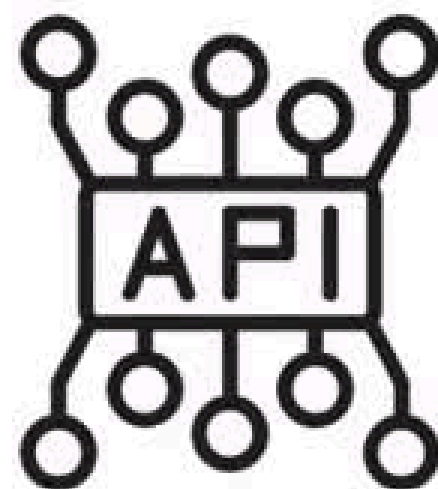
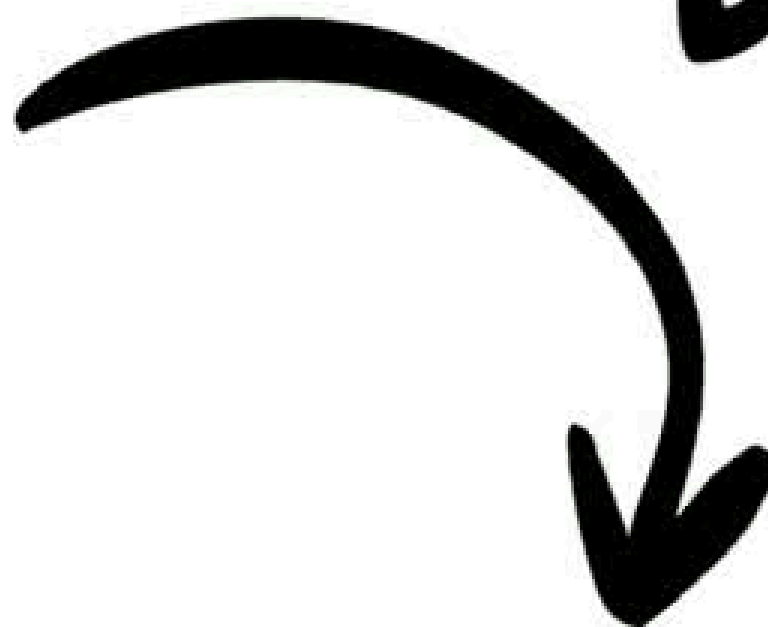
- Feedback-driven retries
- Post-action visual validation
- Ensures errors are caught early

Design Excellence

- True black-box, non-intrusive testing
- Human-like decision loop
- Modular, adaptable, scalable
- Future integration with tools like Jira
- Exploratory testing



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Title of the individual component: Visual Perception

Specialization: Software Engineering

IT22099518 | Hemapriya H.A.N.S



Individual Title: Visual Perception

1. Sub Problems

- Internal screen capturing is restricted
- System cannot understand on-screen content
- Manual screen interpretation is inefficient

2. Prototype Outcomes

- Successful detection of UI components
- Screen data converted into structured format
- Integration with Agentic AI via APIs

Our Solution

- External webcam-based screen capture
- Object detection for UI elements (YOLO)
- OCR for text extraction
- Structured JSON output for Agentic Core

Individual Title: Visual Perception

3. User Requirements Addressed by the Solution

- Automatic screen understanding
- Minimal user interaction
- Real-time visual processing
- Support for accessibility and automation

4. Design Excellence / Individual Contribution

- Modular vision processing pipeline
- API-based component integration

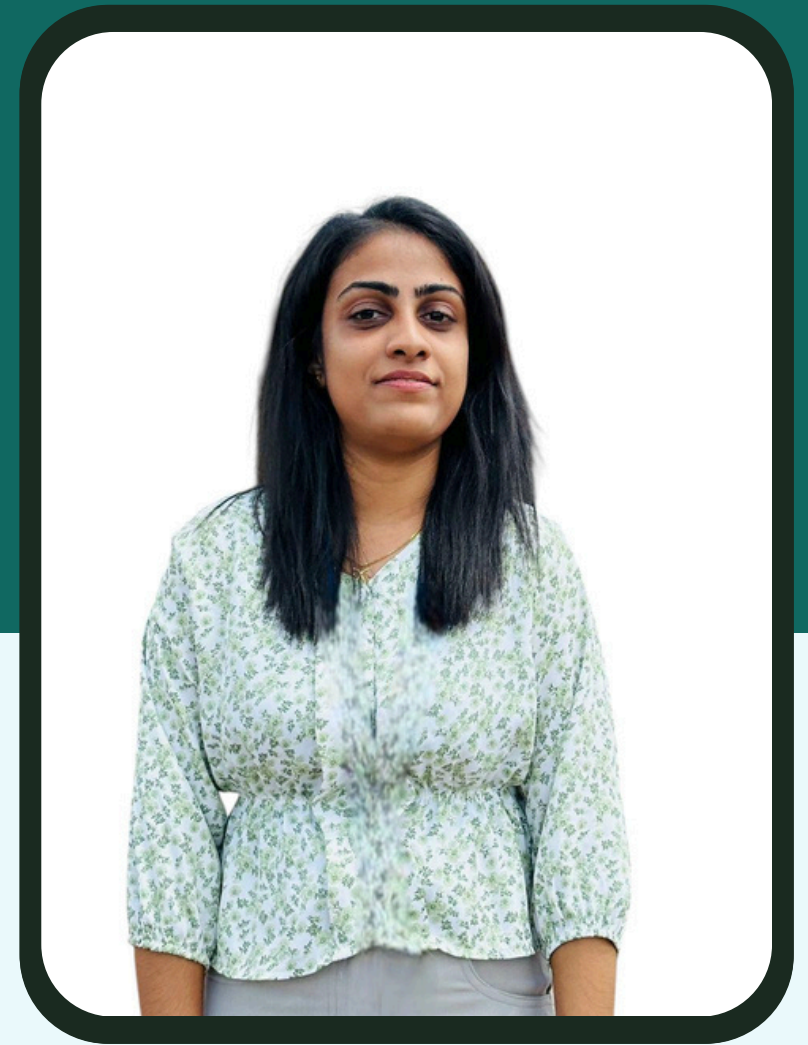
5. User Feedback on Prototype

- UI elements detected accurately
- Improves automation experience
- OCR accuracy needs enhancement

Title of the individual component: LLM & Prompt Engineering

Specialization: Software Engineering

IT22083296 | E.K.K.Tharushi Nethmini Edirisinghe



Individual Title: LLM & Prompt Engineering

1. Sub Problems

- Automation lacks human reasoning
- Poor at exploratory testing

Our Solution

- LLM + RAG agent generates step-by-step actions
- Understands intent, adapts dynamically

2. Prototype Outcomes

- Successfully generates test steps for:
 - General software functionalities
 - UI-based interactions

3. User Requirements Addressed by the Solution

- Convert natural language instructions into test steps
- Support functional & exploratory testing
- Provide domain-specific knowledge via RAG

Individual Title: LLM & Prompt Engineering

4. Design Excellence / Individual Contribution

- Fine-tuned LLM for task-specific actions
- Prompts for step sequencing & human-like decisions
- Hybrid Architecture: general model + RAG

5. User Feedback on Prototype

- Steps clear and easy to follow
- Better exploratory behavior vs rule-based automation
- Good balance of model & retrieval reasoning

Title of the individual component: HID Actuation

Specialization: Software Engineering

IT22167378 | Hasintha H.I.G.A



Individual Title: HID Actuation

1. Sub Problems

- No safe method to execute AI actions in black-box environments
- Software automation tools are OS-dependent and detectable
- Lack of state awareness of physical HID device
- No synchronization between decisions and device execution state

2. Prototype Outcomes

- Successful keyboard and mouse HID emulation via ESP32
- Accurate execution of AI-generated input commands
- Achieved human-like input timing and behavior
- Enabled black-box UI interaction without OS access
- Device Shadow successfully mirrors the HID state

Our Solution

- ESP32-based HID emulation (keyboard & mouse)
- Device Shadow to represent real-time HID state
- Device Shadow maintains connection status, current action state, and execution acknowledgements
- Enables feedback-driven and recoverable execution

3. User Requirements Addressed by the Solution

- Works on any operating system without drivers
- Real-time execution status visibility via Device Shadow
- Supports adaptive actions based on feedback
- High reliability for long-running automation tasks

Individual Title: HID Actuation

4. Design Excellence / Individual Contribution

- Hardware-level automation instead of software hooks
- Hardware-level automation over software hooks
- Modular HID command protocol with acknowledgements
- Device shadow- Transport layer, Motion engine, Command Queue, Shadow state manager, HID interface
- ESP 32 - HID interface, Command processor, Feedback generator

5. User Feedback on Prototype

- Inputs perceived as human-like
- Successful cross-platform operation
- Device Shadow improved debugging and reliability
- Requests for advanced gesture support

Title of the individual component: Agentic AI Logic

Specialization: Software Engineering

IT22125248 | Annesiyani Srikanthan



Individual Title: Agentic AI Logic

1. Sub Problems

- Manual UI testing is time-consuming
- Automation fails on black-box systems
- No visual feedback verification
- UI changes break fixed scripts

2. Prototype Outcomes

- ✓ Closed-loop automation
- ✓ Screen → decision → action flow
- ✓ Basic retry and self-correction
- ✓ Action, decision, and result logging
- → Planned: automated report generation
- → Planned: Jira integration for QA reporting

Our Solution

- Agentic Perceive–Plan–Act–Evaluate loop
- Uses screen OCR as system state
- LLM generates adaptive actions
- Visual feedback validates results

3. User Requirements Addressed by the Solution

- No APIs or system access needed
- Works on any visible UI
- Natural language task input
- Reduced manual QA effort

Individual Title: Agentic AI Logic

4. Design Excellence / Individual Contribution

- Human-like agentic decision loop
- Screen-based automation (no internal access)
- Adaptive LLM-driven action planning
- Feedback-driven retries and validation
- Modular design for easy extension (HID, Jira)

5. User Feedback on Prototype

- Reduces repetitive manual QA work
- Easy to use via natural language input
- Useful for testing legacy / restricted systems
- Strong potential for real-world QA automation

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Market Size

- Growing global software testing & QA automation market
- High demand from enterprises with legacy and regulated systems

What is Unique in our Solution

- Camera-based, screen-aware automation
- No APIs, no code access, no installation required
- Human-like agentic AI decision loop
- Works on any visible UI

Commercialization / Sustainability

- Offered as a subscription-based QA automation tool
- Scalable across multiple projects and teams
- Integrates with existing QA workflows (e.g., Jira)

Cost Recovery Model

- Monthly or yearly subscription plans
- Enterprise licensing for large QA teams
- Reduced manual testing cost offsets usage cost

Customer Persona

KAMAL PERERA



AGE : 30
GENDER : Male
LOCATION : 123/1, Malabe

ABOUT

Kamal Perera is a junior QA engineer who struggles with repetitive manual testing and time-consuming exploratory tasks. He seeks a solution that automates testing in a human-like way, handles both software and hardware interactions, and allows him to test efficiently without writing complex scripts.

GOALS

- Reduce manual testing effort
- Perform human-like exploratory testing
- Test applications without writing complex scripts
- Quickly validate UI changes and new features
- Test both software and hardware-integrated systems

SKILLS

- Manual testing and test case execution
- Basic knowledge of automation tools (e.g., Selenium, Appium)
- Understanding of software development life cycle (SDLC)
- Problem-solving and analytical thinking

PERSONALITY

- Curious and detail-oriented
- Patient but sometimes frustrated with repetitive tasks
- Methodical and logical
- Open to learning new tools and technologies
- Prefers structured guidance

FAVOURITE BRAND





*Thank
You*

